

Quiz 3 - Part 1: Discrete Mathematics (CSE121)

Date: 13-10-2025

Maximum marks: 30

Instructions. Please read the following points carefully.

- Mention your names, roll numbers, room number, and the part number of the quiz in your answer sheets. There will be a penalty associated with missing any of these details.
- Do not use your mobile phones, tablets, laptops, books, notebooks, or any other material during the exam.
- If you are found copying, the matter will be immediately reported to the Disciplinary Action Committee (DAC). If you have any questions, please ask the instructor or TAs.
- This exam is designed for 50 minutes.

1 Problem Set

Question 1.1. 1. (4 marks). Let X, Y be arbitrary non-empty sets. Prove or disprove the following: every surjective function $F : X \rightarrow Y$ is also injective.

2. (3 marks). How many minimum numbers should we pick from 1 to 100 to guarantee that there exist two numbers in the chosen list whose difference is at most 1? Justify your answer.
3. (3 marks). Let A, B be subsets of the universal set U . Prove that $\overline{A \cap B} = \overline{A} \cup \overline{B}$ by using only the rules of propositional logic (given in the Appendix), where for any $S \subseteq U$, $\overline{S} := U - S$.

Question 1.2. 1. (4 marks). Let $\mathbb{Q}$ be the set of integers and $F : \mathbb{Q} \rightarrow \mathbb{Q}$ be defined as follows: for every $x \in \mathbb{Q}$,

$$F(x) = \begin{cases} 2x + 1 & \text{if } x \geq 0 \\ 2x - 1 & \text{if } x < 0 \end{cases}$$

Prove that F is a function. Is F a bijective function? Prove your answer.

2. (2.5 marks). Let X, Y be arbitrary sets. Prove or disprove: $B \subseteq A$ if and only if $\text{pow}(B) \subseteq \text{pow}(A)$, where $\text{pow}(A)$ is the power set of A .
3. (3.5 marks). Let A, B be non-empty sets and $X \subseteq A, Y \subseteq A$. Let $F : A \rightarrow B$ be an arbitrary function. For any subset $Z \subseteq A$, $F(Z) := \{f(a) \in B : a \in Z\}$. Prove or disprove: $F(X \cup Y) = F(X) \cup F(Y)$.

- Question 1.3.** 1. (5 marks). Let A, B, C be non-empty sets. Let $F : A \rightarrow B$, $G : C \rightarrow A$, and $H : C \rightarrow A$ be functions. Prove or disprove: F is injective if and only if the following holds: if $F \circ G = F \circ H$ then $G = H$. Here, $=$ means the equality of functions.
2. (2 marks). Describe the power set of $\{1, 2, 3\}$.
3. (3 marks). Let $F : \mathbb{Z} \rightarrow \mathbb{Z}$ be such that for every $x \in \mathbb{Z}$, $F(x) := x + 3$. Is $\text{range}(F)$ a proper subset of $\mathbb{Z}$, where $\mathbb{Z}$ is the set of integers? Prove your answer.

A Rules of the algebra of propositional logic

Let p, q, r be propositional variables, and T, F denote true and false. The rules are stated below.

1. Rules for AND.

- (a) **Commutativity of AND:** $p \wedge q \equiv q \wedge p$.
- (b) **Associativity of AND:** $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
- (c) **Identity for AND:** $T \wedge p \equiv p$.
- (d) **Falsity for AND:** $F \wedge p \equiv p$.
- (e) **Idempotence for AND:** $p \wedge p \equiv p$.
- (f) **Contradiction for AND:** $p \wedge \neg p \equiv F$.

2. Rules for OR.

- (a) **Commutativity of OR:** $p \vee q \equiv q \vee p$.
- (b) **Associativity of OR:** $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
- (c) **Identity for OR:** $F \vee p \equiv p$.
- (d) **Truth for OR:** $T \vee p \equiv T$.
- (e) **Idempotence for OR:** $p \vee p \equiv p$.
- (f) **Validity for OR:** $p \vee \neg p \equiv T$.

3. Other rules.

- (a) **Double negation:** $\neg(\neg p) \equiv p$.
- (b) **Distributivity of AND over OR:** $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
- (c) **Distributivity of OR over AND:** $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- (d) **DeMorgan's laws:** $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.

Quiz 3 - Part 2: Discrete Mathematics (CSE121)

Date: 13-10-2025

Maximum marks: 30

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- This exam is designed for 50 minutes.

1 Problem Set

Question 1.1. 1. (4 marks). Let X, Y be arbitrary non-empty sets. Prove or disprove the following: every injective function $F : X \rightarrow Y$ is also surjective.

2. (3 marks). Let X, Y, Z be non-empty finite sets such that Y contains m rational numbers, Z has n irrational numbers, and X contains integers. Let $F : X \rightarrow Y \times Z$ be an arbitrary function. How many elements should X contain to guarantee that F is not injective? Justify your answer.

3. (3 marks). Let A, B be subsets of the universal set U . Prove that $\overline{A \cup B} = \overline{A} \cap \overline{B}$ by using only the rules of propositional logic (given in the Appendix), where for any $S \subseteq U$, $\overline{S} := U - S$.

Question 1.2. 1. (4 marks). Let $\mathbb{Z}$ be the set of integers and $F : \mathbb{Z} \rightarrow \mathbb{Z}$ be defined as follows: for every $x \in \mathbb{Z}$,

$$F(x) = \begin{cases} x + 3 & \text{if } x \text{ is odd} \\ x - 3 & \text{if } x \text{ is even} \end{cases}$$

Prove that F is a function. Is F a bijective function? Prove your answer.

2. (2.5 marks). Let X, Y be arbitrary sets. Prove or disprove: $A \subsetneq B$ if and only if $\text{pow}(A) \subsetneq \text{pow}(B)$, where $\text{pow}(A)$ is the power set of A . Here, $\subsetneq$ means proper subset.

3. (3.5 marks). Let A, B be non-empty sets and $X \subseteq A, Y \subseteq A$. Let $F : A \rightarrow B$ be an arbitrary function. For any subset $Z \subseteq A, F(Z) := \{f(a) \in B : a \in Z\}$. Prove or disprove: $F(X \cap Y) = F(X) \cap F(Y)$.

Question 1.3. 1. (5 marks). Let A, B be non-empty sets, and $F : A \rightarrow B$ and $G : B \rightarrow A$ be functions such that $B \circ F = I_A$, where I_A is the identity function on A , i.e., for every $x \in A, I_A(x) = x$. What can you say about the injectivity and the surjectivity of F and G ? Justify your answer.

2. (2 marks). Prove that the empty set is a subset of every set.

3. (3 marks). Let $F : \mathbb{Q} \rightarrow \mathbb{Q}$ be such that for every $x \in \mathbb{Q}, F(x) := 2x - 1$. Is $\text{range}(F)$ a proper subset of $\mathbb{Q}$, where $\mathbb{Q}$ is the set of rational numbers? Prove your answer.

A Rules of the algebra of propositional logic

Let p, q, r be propositional variables, and T, F denote true and false. The rules are stated below.

1. Rules for AND.

- (a) Commutativity of AND: $p \wedge q \equiv q \wedge p$.
- (b) Associativity of AND: $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$.
- (c) Identity for AND: $T \wedge p \equiv p$.
- (d) Falsity for AND: $F \wedge p \equiv F$.
- (e) Idempotence for AND: $p \wedge p \equiv p$.
- (f) Contradiction for AND: $p \wedge \neg p \equiv F$.

2. Rules for OR.

- (a) Commutativity of OR: $p \vee q \equiv q \vee p$.
- (b) Associativity of OR: $(p \vee q) \vee r \equiv p \vee (q \vee r)$.
- (c) Identity for OR: $F \vee p \equiv p$.
- (d) Truth for OR: $T \vee p \equiv T$.
- (e) Idempotence for OR: $p \vee p \equiv p$.
- (f) Validity for OR: $p \vee \neg p \equiv T$.

3. Other rules.

- (a) Double negation: $\neg(\neg p) \equiv p$.
- (b) Distributivity of AND over OR: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$.
- (c) Distributivity of OR over AND: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$.
- (d) DeMorgan's laws: $\neg(p \wedge q) \equiv \neg p \vee \neg q$ and $\neg(p \vee q) \equiv \neg p \wedge \neg q$.